

H5 — ML + Embedded Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: any member can run a training run on Colab and survive a disconnect.

Steps: 1. Run a 50-step probe on Colab: `uv run python -m research.tinystories.train --arm baseline --vocab 32768 --steps 50 --target-core 559000 --seed 0` 2. Find where train.py saves checkpoints. Move them to Drive. 3. Kill the session. Reopen it. Resume from the checkpoint. 4. Write the Colab runbook: keep-alive, Drive save, resume, quota notes.

Deliverable: RUNBOOK.md in the club repo.

Verify: a killed session resumes from the last checkpoint without restarting from step 0.

Learning this week

Neural network basics. Watch 3Blue1Brown episode 2. Explain what an epoch is.

Resources

- Colab: <https://colab.research.google.com>
- Kaggle (fallback): <https://www.kaggle.com>
- train.py: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/train.py>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Done when

- [] RUNBOOK.md exists in the club repo
- [] A killed session resumes from the checkpoint
- [] You can say what the project does in one sentence